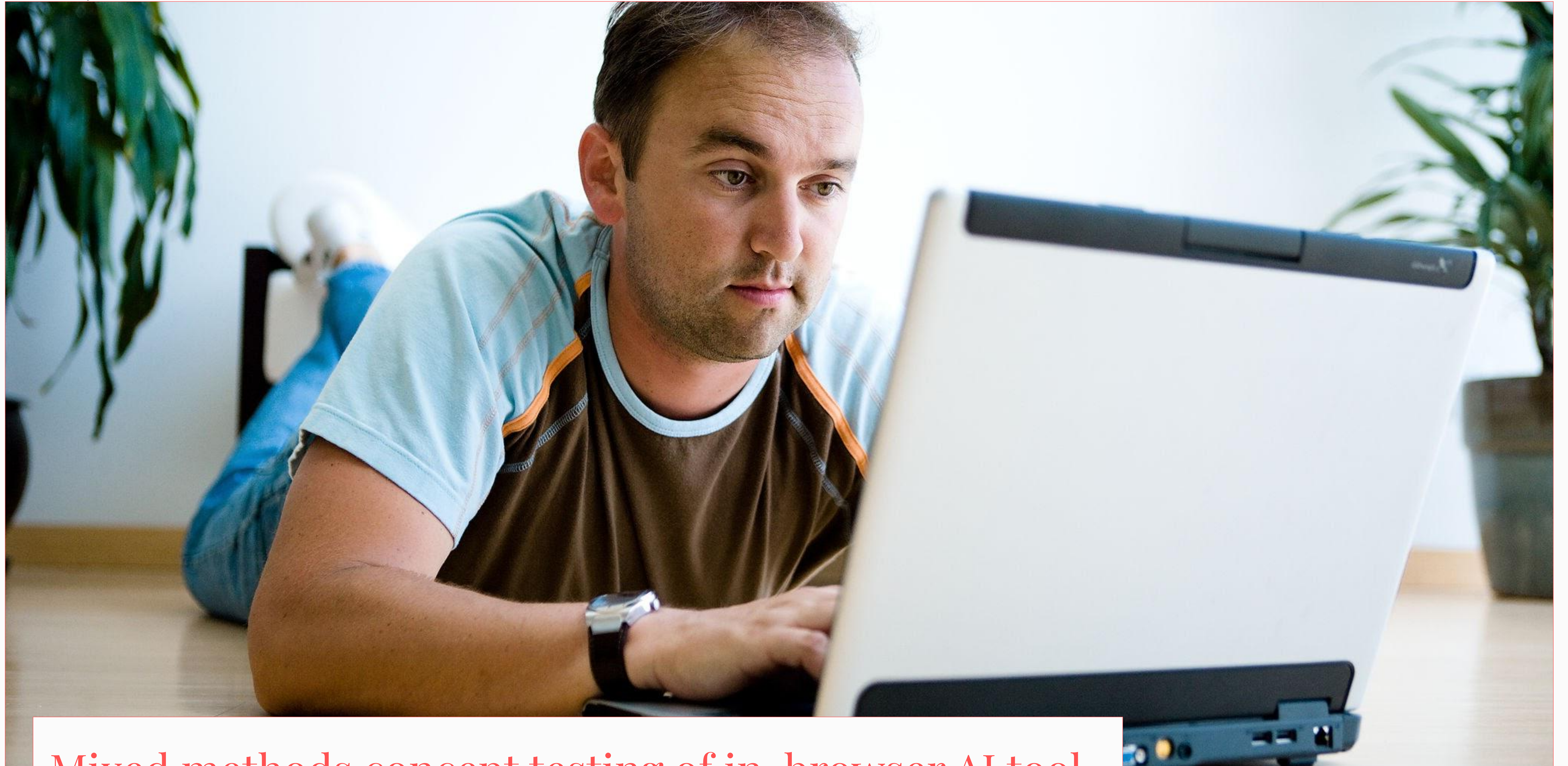




Mixed methods concept testing of in-browser AI tool

Mixed methods concept testing of in-browser AI tool

The client

BetaCo: A Senior Project Manager implementing new in-browser AI tools for a major internet browser for a Fortune 10 tech company

01 Context

The client was anxious about the AI landscape, especially how stand alone Generative AI apps compete with in-browser AI tools, like AI-generated summaries. Given the crowding playing field and novelty of AI solutions, this stakeholder wanted to know *everything* and fast.

02 Issue

How can the browser boost engagement and growth for AI tools in the browsing experience?

03 The ask

Quantitatively and qualitatively...

1. Identify potential opportunities for our AI tool and browser to improve the GenAI + browser experience

2. Explore hypothesized AI tool + browser opportunity and dig sites
3. Determine the direction for integrated AI tool + browser experience that encourages wider adoption



The solution

Instrument: Because of the relative “newness” of the AI-tool’s implementation into the browser, there were few *a priori* assumptions about what pain points and opportunities exist. To that end, we deployed a two-pronged sequential qual-quant attack to obtain a depth of understanding and then to measure it at scale:

Qual*: Sixteen 60-minute online in-depth interviews with AI-tool users

Quant: Fifteen-minute online survey of 5,001 browser/AI-tool users, cut for generation, browser use (e.g., primarily Chrome, Edge, ...), AI tool use (e.g., primarily ChatGPT, Claude, ...), and affinity towards AI.

Timeline: 8 weeks

My role:

1. Client communication (w/ team)
2. Quant survey development
3. Quant data analysis
4. Integration of Qual & Quant results, story-development & visualization (w/ team)

*the following slides only detail my contributions: the quant analysis and integrated report

Aligning on key issues



Client on fishing expedition, c.2025

Client wishlist

The client brought with him a research plan that included a competitive audit, concept testing, and usage analysis among other methods--way too much for a single study.

Through close partnership, I successfully negotiated a much more targeted jobs-to-be-done (JTBD) framework that answered his underlying core question: *What does a browser-based Gen AI tool need to do to succeed?*

Reporting strategy



Pillars & tenants

We tackled this broad question, *What does a browser-based Gen AI tool need to do to succeed?*, in 4 steps, each with their own sub-questions*

*questions below are illustrative only

01

The state of the AI User

- *Which AI tools are users engaging?*
- *How are they engaging them?*
- *Where and for what?*

02

The state of client's product

- *Who is/isn't using our product?*
- *On what devices is our product used?*
- *In what context?*

03

Pain points and opportunities

- *What makes AI tools enjoyable/ difficult to use?*
- *What opportunities are there to improve?*

04

Concept test

- *Will our new AI-in-browser concept keep current users and attract new ones?*

Takeaways from 1,000ft

For the busy Project Manager

With over 5,000 participants from all walks of life, we have a massive amount of data to present. But, this ice-breaker slide gives the busy project manager a sense of what's important:

- 1. Our current strengths
- 2. Our current challenges
- 3. Our next steps

 1 Learning & Discovery At this stage in their AI adoption journey, AI Users need better ways to get educated, especially on what is possible with AI, to increase their confidence and frequency.	 2 From Work to Personal Our strength today is in the Work setting. We will need to encourage more personal use cases with our tools. Consider using a work-personal split to introduce the value of agentic.
 3 The Mobile Experience Our strength today is in the computer format. We will need to improve our user experience on mobile, and help AI Users understand the value of browser embedding over going to an AI App.	 4 Personalize & Customize AI Users are in need of control over how their AI Tools interact with them and the nature of its outputs. We will need to build in settings that give our users this control.

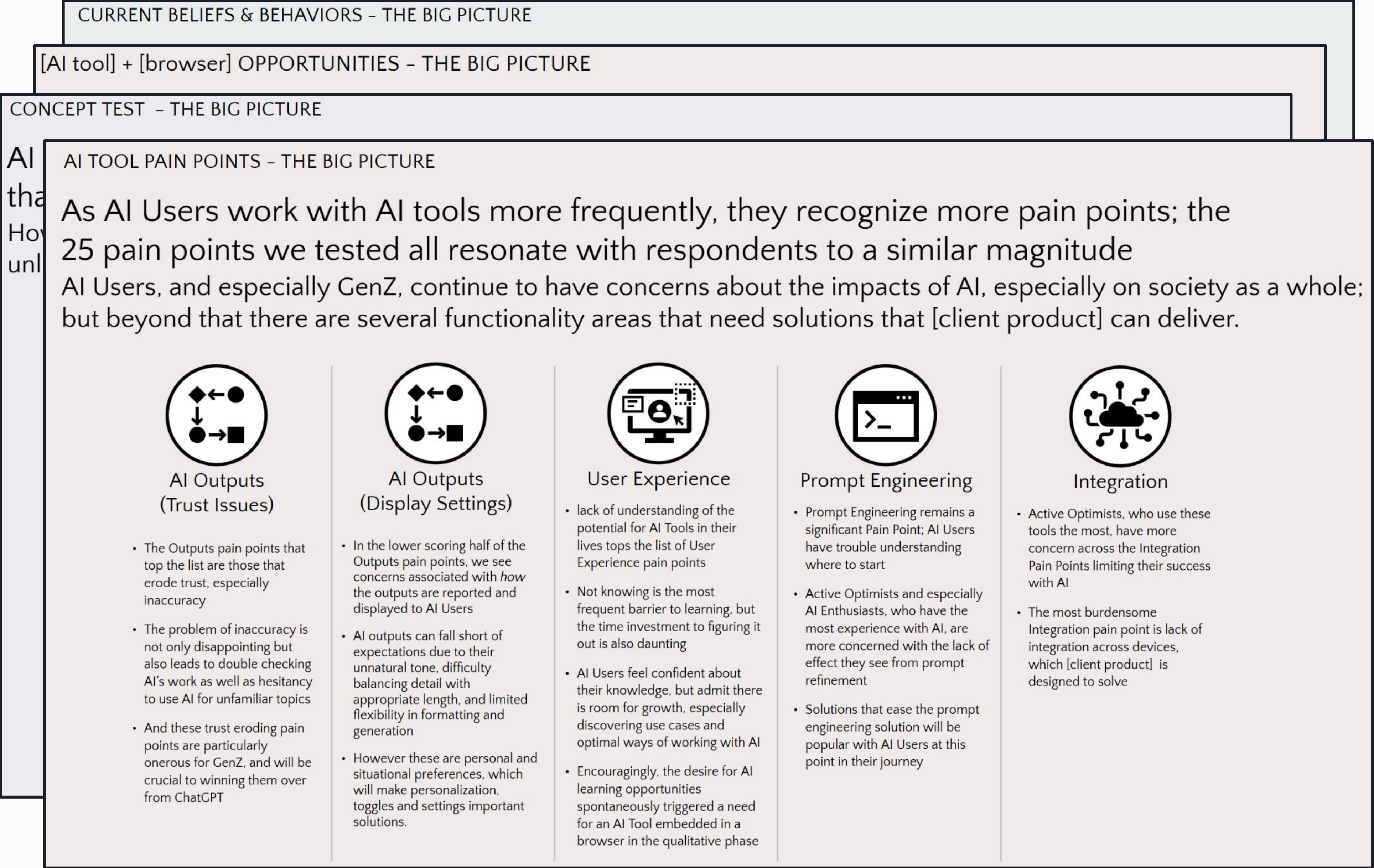
Peeking through the clouds

Pillar summaries

We start gradually adding more depth by summarizing the main takeaways from each pillar.

The summaries not only give a quick look at users, behaviors, and needs, but also identify where the client’s product succeeds and where further work needs to be done.

Through close, iterative partnership with the client, we front-loaded the information most relevant to the product team (design, innovation), senior leadership (strategy), and other interested parties.



AI user manual

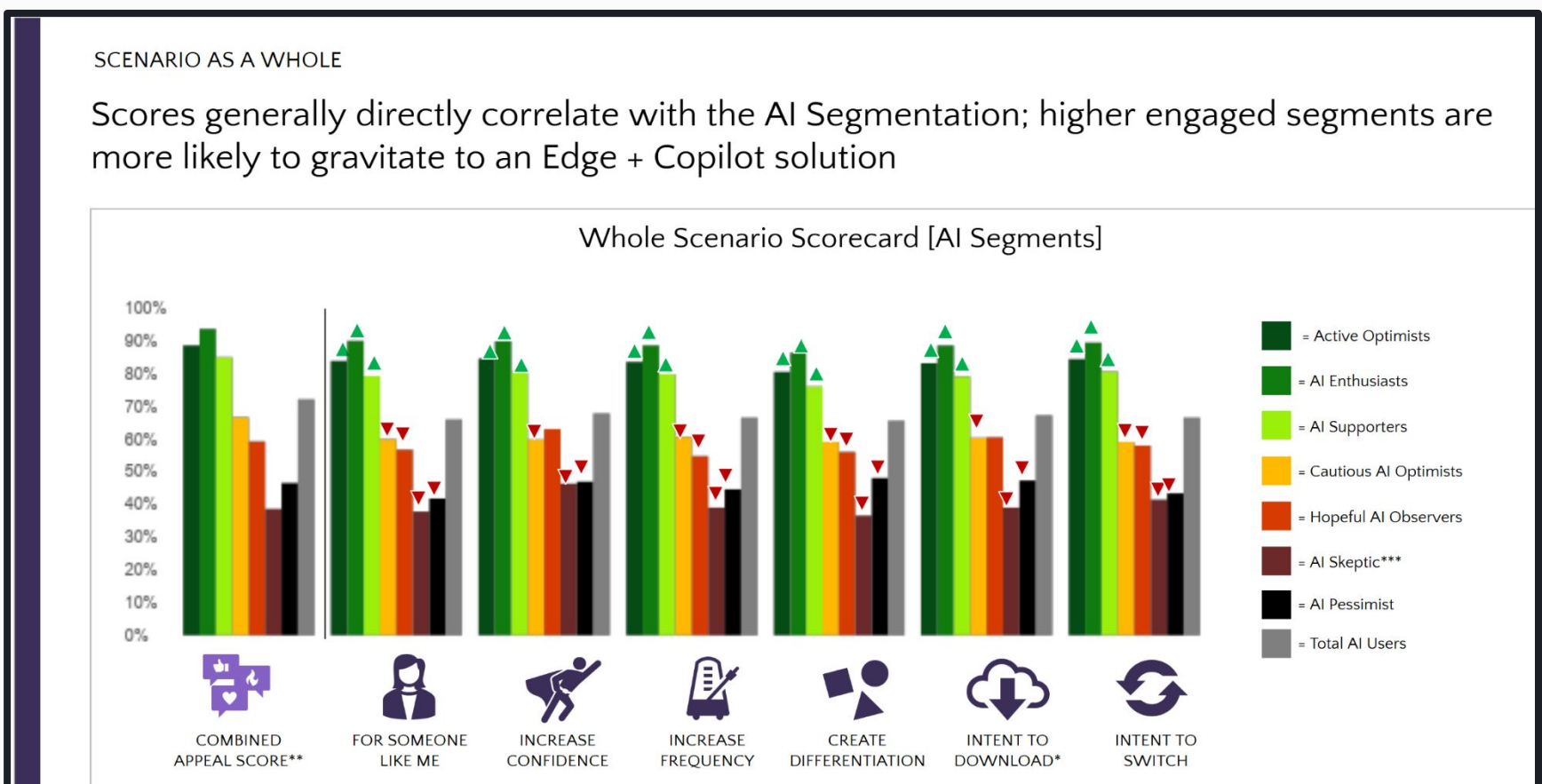
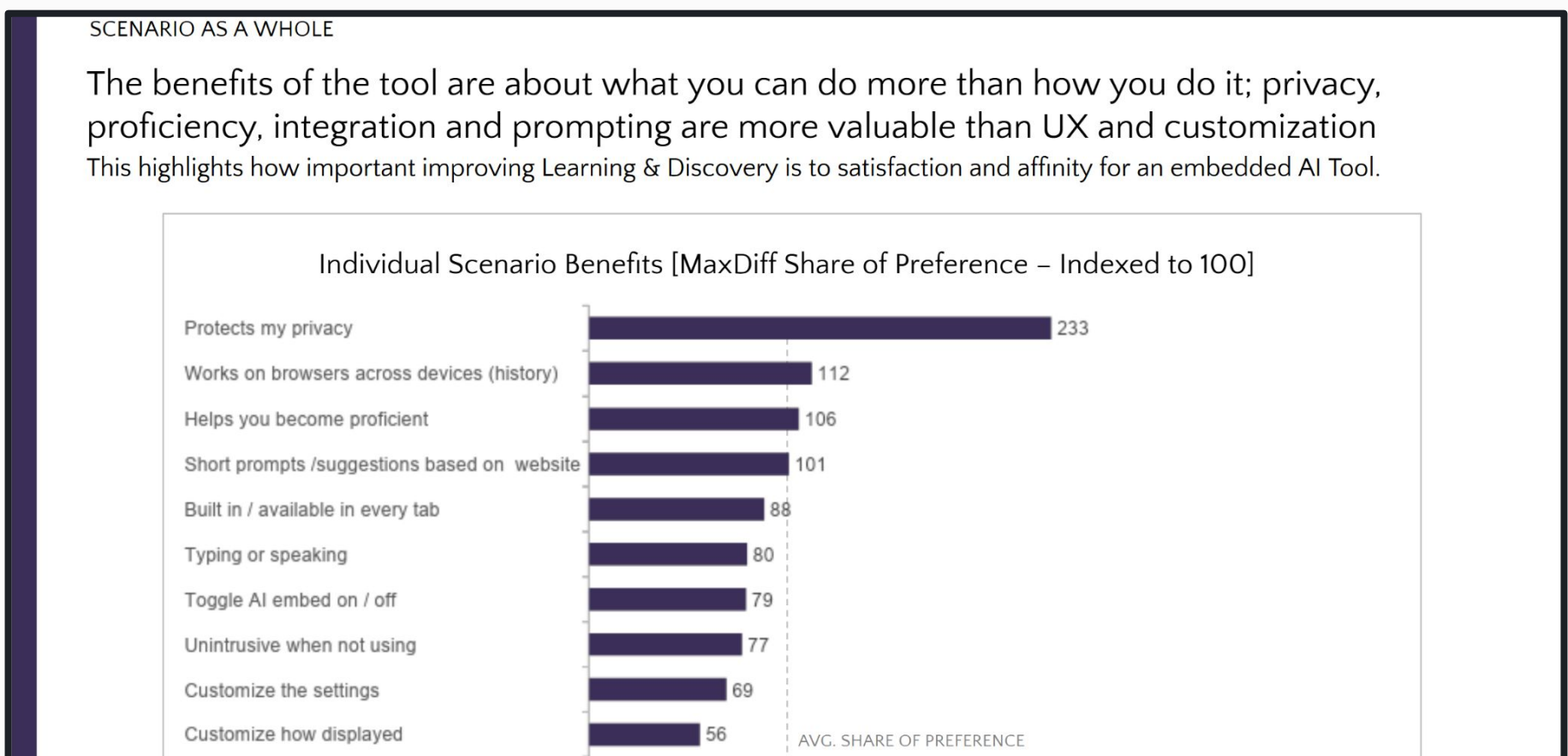
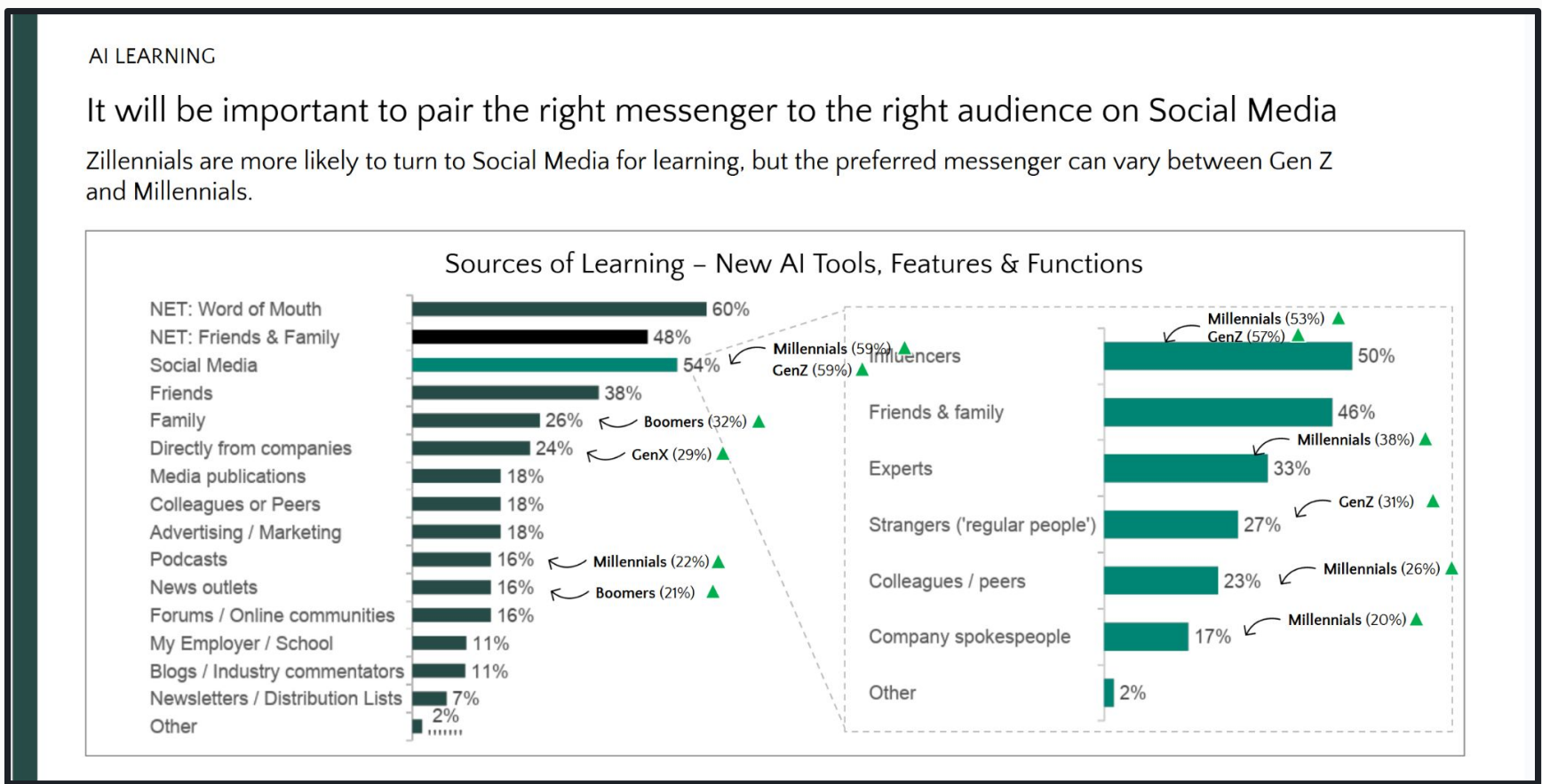
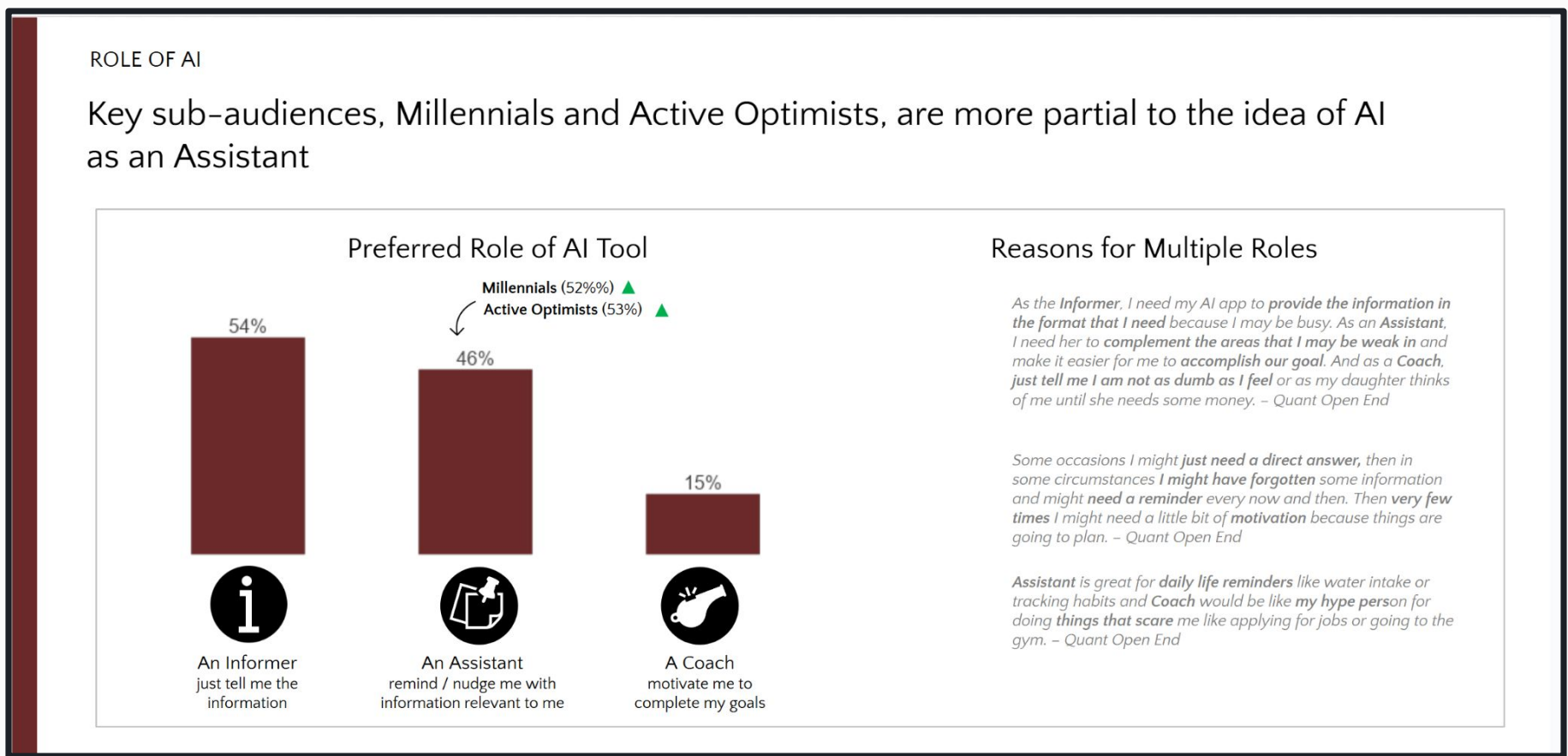
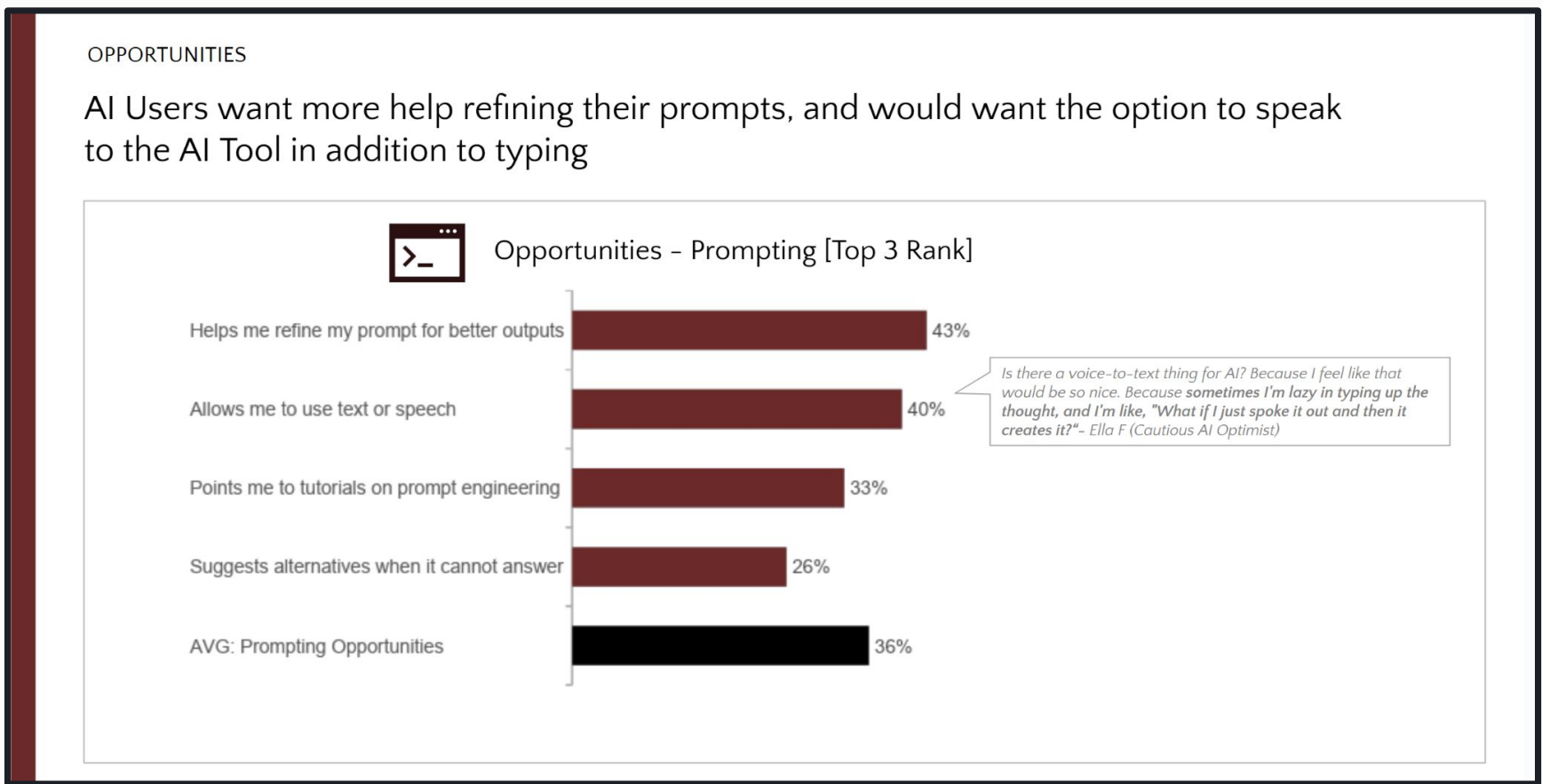
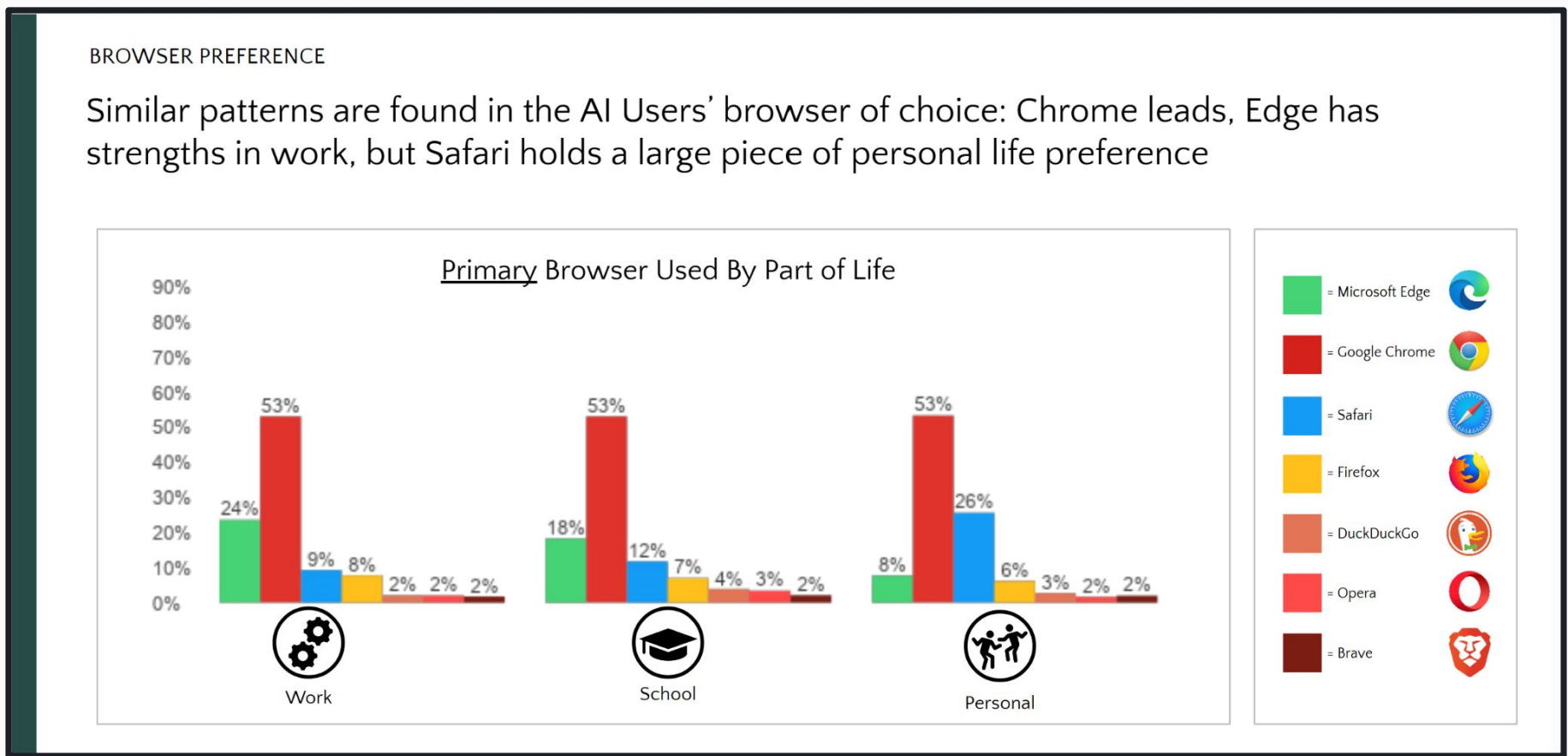
The Official Playbook™

We collected a lot of data from a lot of different users--spanning generations, attitudes towards AI and other behavioral and demographic factors.

Potentially, everything is relevant!

In addition to reporting our top-level findings and recommendations, we prepared a detailed, 150-slide user manual on how to understand the needs, thoughts and behaviors of current and potential AI users.

While my team offered long-tail support on this project, the User Manual allowed the client to self-serve on new questions, grab ready-made slides for product discussions with the broader team, and have the information critical to innovate, refine and launch new products at their fingertips.



Concept test in focus

Pillar 4 was a concept test, which involved one “product” with a few different features to explore.

The next several illustrate our storytelling around the concept as a whole. Then, the final slide in this section shows how one particular feature was evaluated.

(Select) Exercises:

1. Ratings (KPIs)
2. IDIs
3. Highlighter task
4. MaxDiff

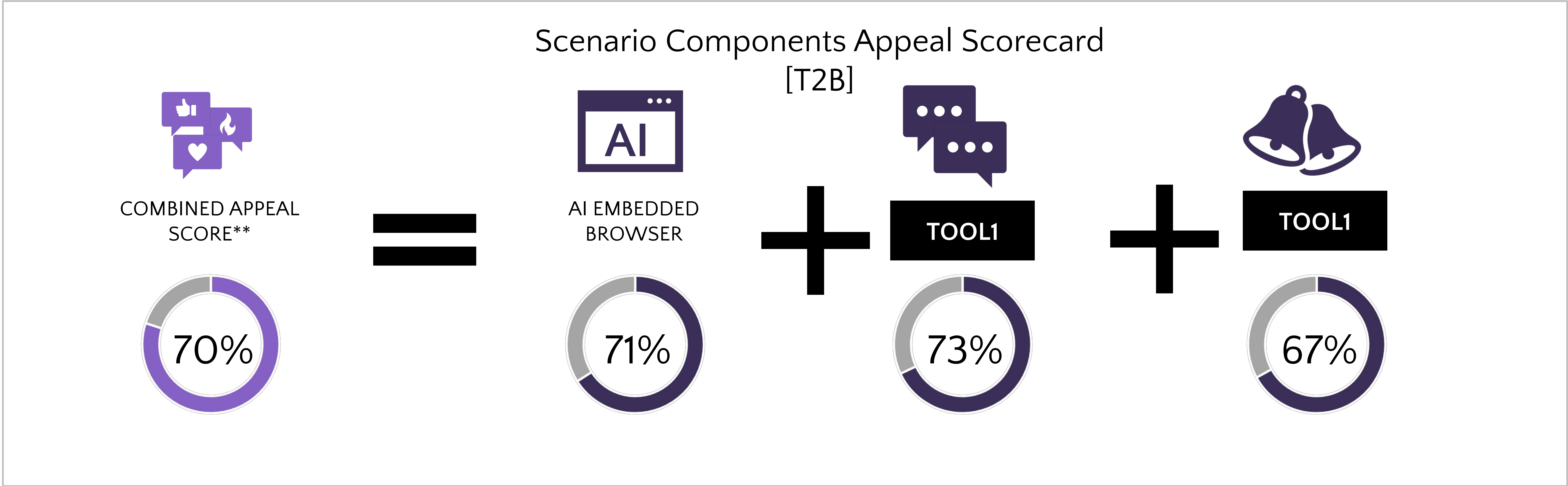


AI Users, especially Millennials give the hypothesized scenario high scores, indicating that this enhancement will support retention and preference

However, Non-AI Users are not there yet; they are much less likely to find the scenario appealing, therefore this is unlikely to be a new-AI User acquisition tool.

THE BROWSER	BROWSER WATCHOUTS	TOOL 1	TOOL 2	PRIORITIZING OPPORTUNITIES
- AI Users agree, embedding an AI Tool in the browser leads to even more 'easier to use' AI - Greater ease of use comes from a combination of benefits tied to embedding: - FEATURE A - FEATURE B - FEATURE C - FEATURE D	- All of this access and integration does raise privacy and security concerns, it will be important to highlight privacy and security toggles, controls and guarantees soon after introducing this scenario to users - Tutorials are somewhat interesting to users, as long as they are short, targeted and if they ask. However, Prompts & Nudges are more interesting learning opportunities that require less effort from the user - Because of browsing and AI habits on phones vs. computers, this feels like a computer solution. We will need to consider how to translate the scenario to a phone-format	- TOOL1 eases initial concerns around an AI embedded browser's potential obtrusiveness, however, some feel that for it to truly be nonintrusive they would need the option to completely disable it for heads-down time - AI Users appreciate the flexibility of being able to reposition the AI chat window, though there is a preference for accessing the AI Assistant on the right side of the webpage - Due to long-time offerings like [competitor], talk is expected, especially on mobile interactions - Tailored suggestions feels more conversational, closer to an assistant than just TOOL1, and they also begin to address some of the pain points around prompt engineering	- TOOL2 goes a long way to improving the Learning & Discovery pain points - However, as with all recommender widgets, the key is balance and relevance in that moment - Each of the TOOL2 examples plays a role in bringing the whole scenario and its benefits to life: - Podcast Illustrates Discovery - Tracking Prices Illustrates Advanced Search - Art Project Illustrates Context	- The benefits of the tool are about what you can do more than how you do it; privacy, proficiency, integration and prompting are more valuable than UX and customization - This highlights how important improving Learning & Discovery is to satisfaction and affinity for an embedded AI Tool - Overall, AI Users are left with a positive impression of the scenario and can easily imagine its value add to their journey of AI discovery and growing their expertise

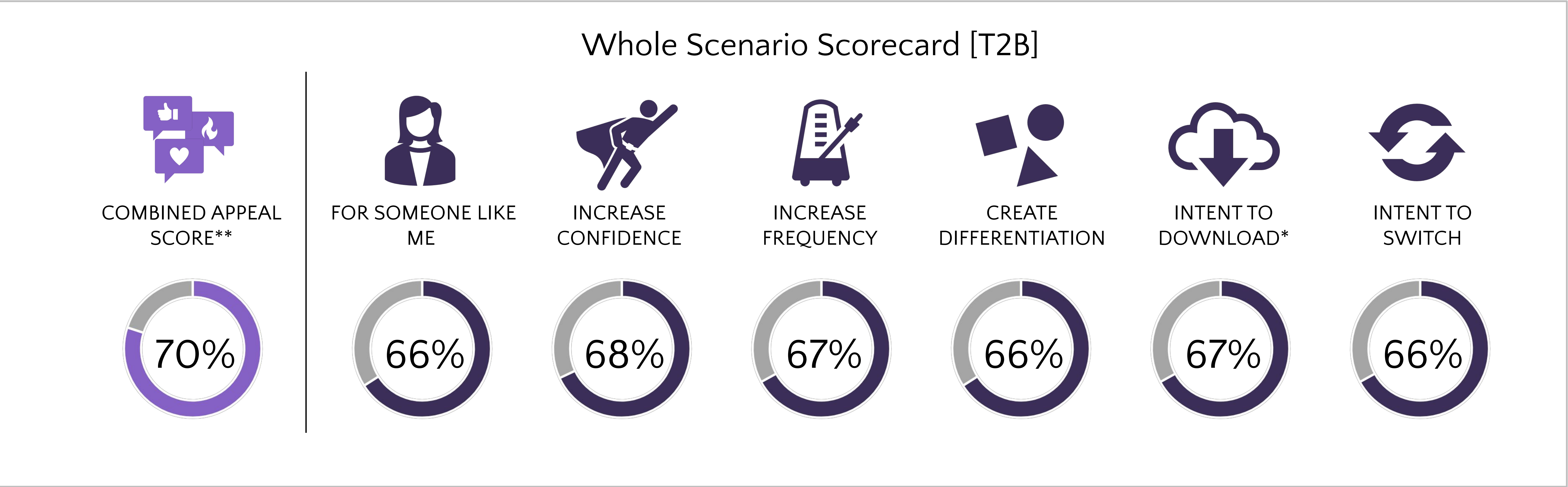
The hypothesized scenario scores strongly on appeal across its component parts



Q: Considering these all together as a single browser idea. Please rate the following statements according to how much you agree or disagree based on the entire idea as a whole.
Please rate this statement according to how much you agree or disagree. The browser is appealing to me
N: AI Users (3,569)
* Not explored in qualitative phase ** Average of all three pieces of stimulus (Appeal of Browser + Appeal of QA + Appeal of P&N)

SCENARIO AS A WHOLE

In fact, AI Users give the scenario high scores across the board, indicating that this enhancement will support retention and preference among AI Users

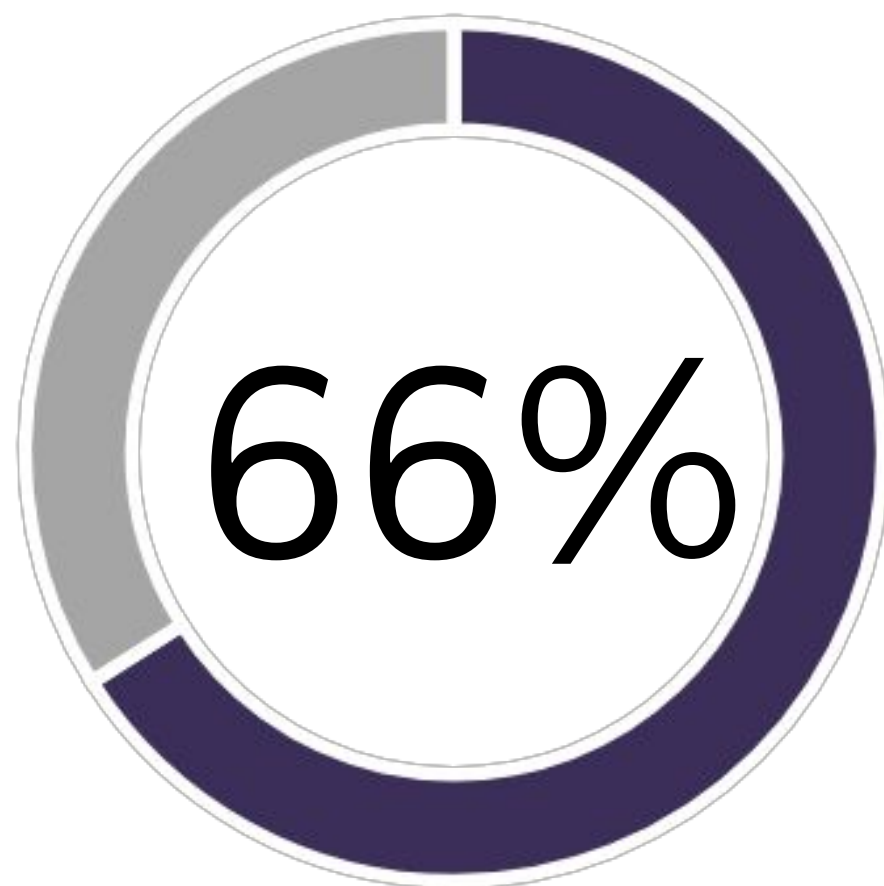


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N: AI Users (3,569)
* Not explored in qualitative phase ** Average of all three pieces of stimulus (Appeal of Browser + Appeal of QA + Appeal of P&N)

Respondents can easily imagine themselves using an AI enabled browser like this one, however it does not feel personalized enough to truly be ‘for them’



FOR SOMEONE
LIKE ME



It fits with their AI in browser needs, and they can see it leading to a more seamless experience while addressing some of the challenges around discovering uses cases.

*That **would solve one of my main struggles now** because I just don't know some of the capability. I was not raised on AI. My brain is not like I'm going to use AI for that.*
– Rachel Z (AI Enthusiast)

*This is the **perfect AI tool for me**. I think it's going to be very **helpful in my future endeavors on the internet and the browser**.*
– Dee J (AI Enthusiast)

*I'm **all for simplicity, productivity, efficiency**, which I think this is exactly what it can do.* – Gene K (Cautious AI Optimist)

However, it feels like it would be just as helpful to anyone, lacking features that convey a more bespoke or personalized experience.

*I think anyone could use it. I don't think it's just **made for me**? I think anyone could pick up and use it.* – Christy S (Cautious AI Optimist)

*I would probably just say it's a 3 [out of 10]. Because if **it's for me or if it's bespoke or if it's something custom**, I feel like I would ask for it first, but I didn't ask for it, so **I'm not sure if it's exactly for me**.*
– Jack L (AI Enthusiast)

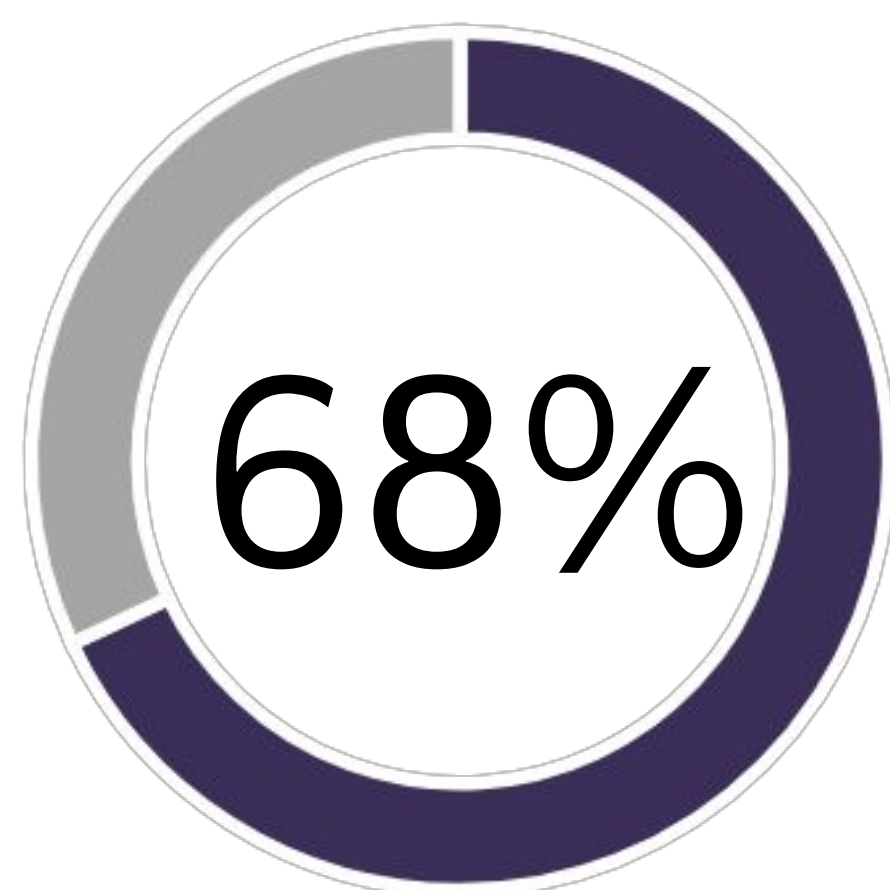


(click to play video, transcript in slide notes)

Reactions to it increasing their confidence depends on their confidence before seeing the scenario; they're either looking for more ways to use AI or are already satisfied



INCREASE
CONFIDENCE



Some imagine that the discoverability that this scenario offers will naturally increase their confidence as they uncover more use cases and applications.

It will help show you and help you discover different ways that it can be used, which would help me feel more confident with using it. It would help you figure out what you don't know because you don't know it, but it knows it. – Rachel Z (AI Enthusiast)

Basically, I was like, "Wow, well, if it made those kind of suggestions, it makes me want to think about using AI more for the same reasons." Maybe it would give me other suggestions for other ideas. – Aaron L (Cautious AI Optimist)

Knowing that AI is there as a support and friend would make me feel even more confident. And having it embedded in the browser when I'm doing my daily searches would increase my confidence as well. – Dee J (AI Enthusiast)

Others feel they are already so confident in their AI skills that an AI embedded browser would do little to bolster this confidence even further.

I don't feel like I'm bad at using AI because it's really just asking it questions. I don't think this would make me feel like, 'Oh, I'm better at using AI.' – Landon B (Cautious AI Optimist)

I'm fairly confident. This doesn't change it one way or another. I'm confident I would be able to figure out if I needed to which prompts to use eventually. – Gene K (Cautious AI Optimist)

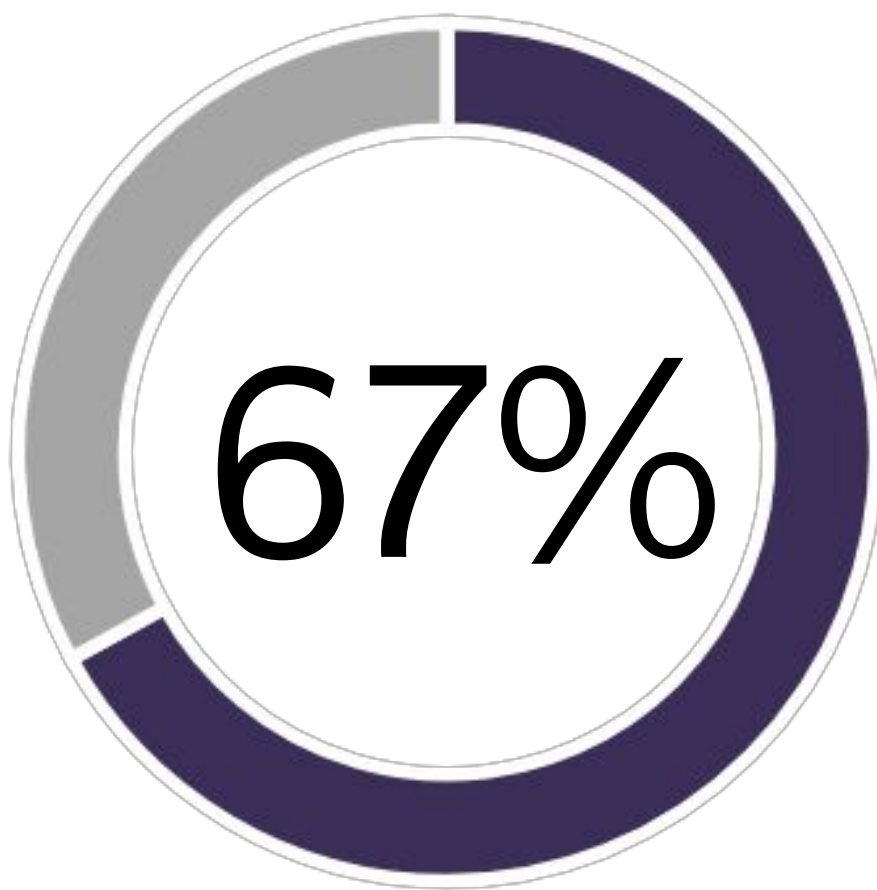
I'm pretty confident, since I already know about it, I guess it's just a different way to use it. – Amelia V (Cautious AI Optimist)

I don't think it personally it increases my confidence at all. But I can see that the question is probably just catered towards someone who doesn't use AI too often. – Jack L (AI Enthusiast)

Respondents agree that having an AI that’s embedded in a browser and surfaces suggestions or prompts would increase their AI in browser usage



INCREASE
FREQUENCY



Having the icon present on any webpage they visit without having to open another tab will make using AI while browsing more top of mind and convenient.

- When it's staring at your face, you click. It will just give me the summary of everything I need to know in a step-by-step guide.* – Gene K (Cautious AI Optimist)

This would increase the frequency I use AI because it's embedded in the browser and it's even more helpful than before. – Dee J (AI Enthusiast)
- It's trying to make it more convenient to be able to use it and not have feel like, oh, here's the tab with the AI.* – Landon B (Cautious AI Optimist)

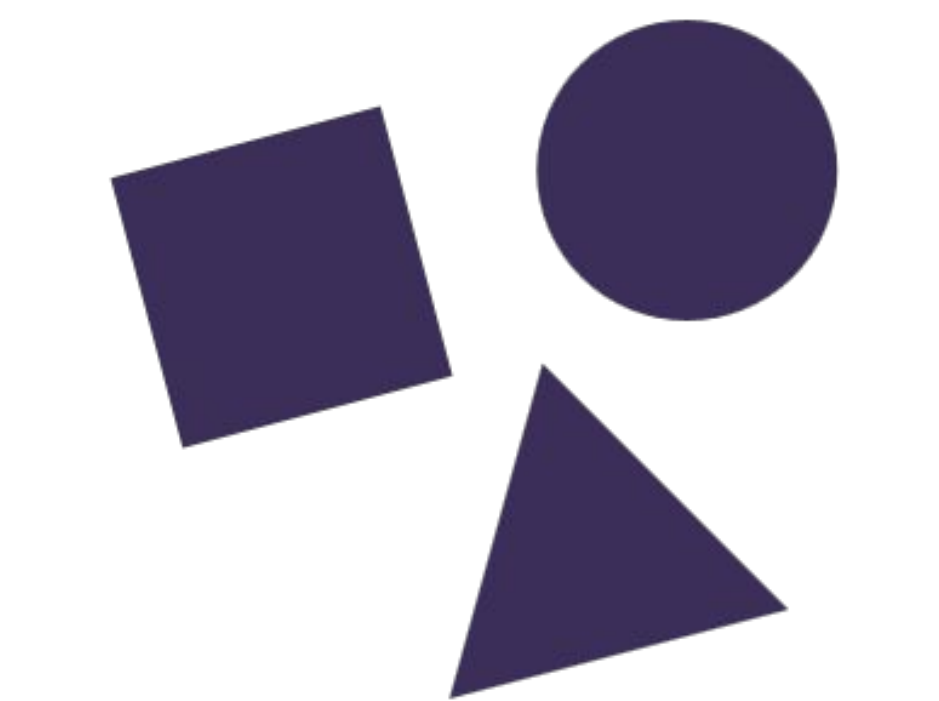
It's like playing basketball with taller shoes. The aid is that everything's in the same place, it's going to push me to use it more. – Eduardo D (AI Enthusiast)

And they imagine the suggestions will add additional use cases to the ones that they are already doing increasing their breadth of usage.

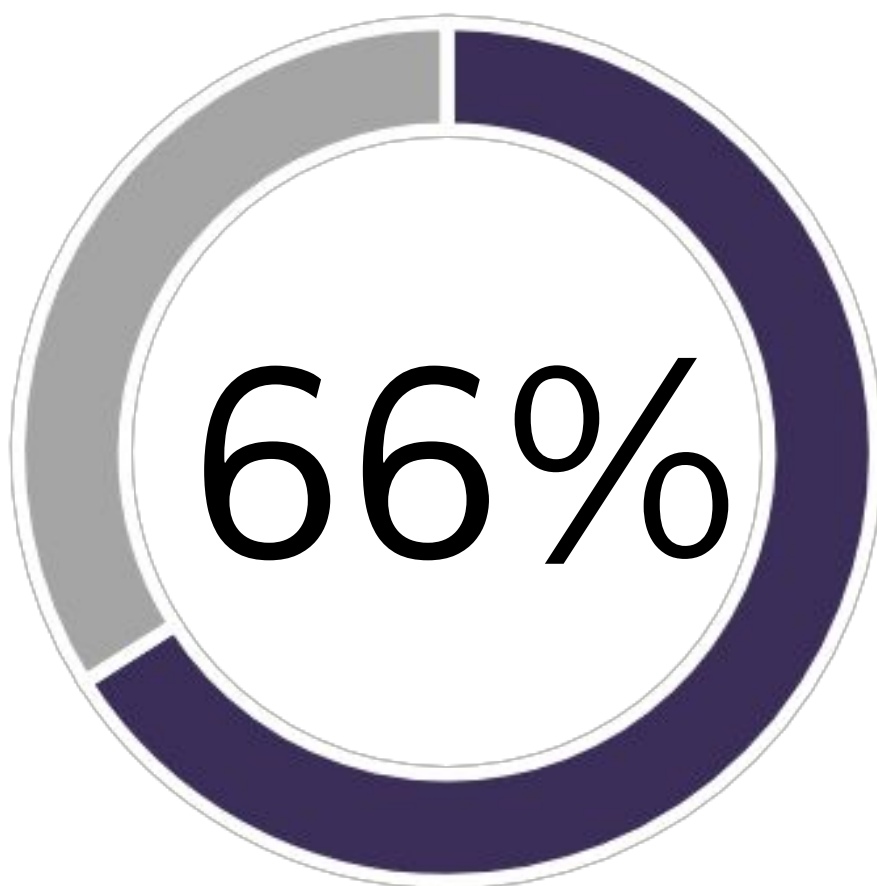
- The more that I learn about it and the more that it can do, the more I'm going to use it. It's offering me more suggestions, and it's giving those little prompts. It's like "think about this."* – Christy S (Cautious AI Optimist)

It would be suggesting things in real time. So, it would increase the frequency because then it would be suggesting ways that I wouldn't even have used it for previously. I'd still be using it for the ways I currently am and then in addition to things that it's suggesting. – Rachel Z (AI Enthusiast)
- Some of the suggestions would be new to me and I'm sure there would be some that are pretty useful that I wouldn't have thought of. So, I would use it in a new way and probably more frequently.* – Peter S (AI Enthusiast)

Although not everyone is sure if AI enabled browser currently exist, consensus is that it will soon be tablestakes with every browser following suit



CREATE
DIFFERENTIATION



Many admit that they don't have a lot of experience in the realm of AI enabled browsers but imagine there are similar tools already out there or coming soon.

*I know that **some of these things** are out there. I don't know how new and different **some of it is**.*
- Christy S (Cautious AI Optimist)

*It's **not something** that I'm familiar with using and having used before, **but it doesn't mean it doesn't exist**.*
- Rachel Z (AI Enthusiast)

I think I've seen it before. But I haven't seen it too often. - Jack L (AI Enthusiast)

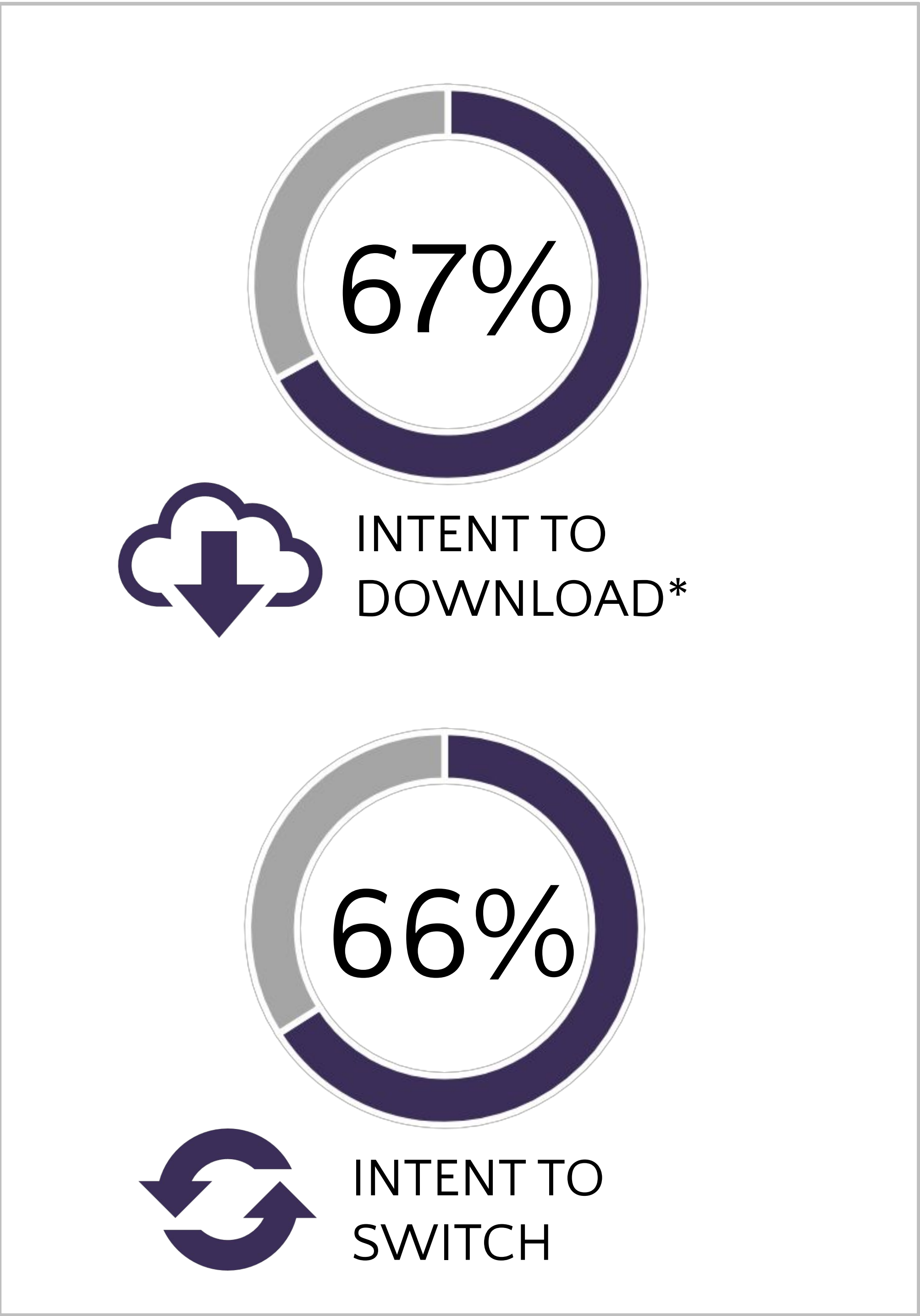
*It is different than what other AI Tools are offering, but I could also see Google Gemini snatching the idea and trying to do **something very similar** as well. So, **while it is new**, I can see other companies trying to replicate.*
- Dee J (AI Enthusiast)

A few feel it's similar to what Google is already starting to do with Gemini.

*I think Google and others, they do offer something similar with the whole AI overview, they also have AI mode. But it almost sounds **like it's describing what's there today**. I'm not sure if I'm seeing any breakthroughs or anything new- Gene K (Cautious AI Optimist)*

With Chrome, Gemini is all over the web pages. You just got to look for the little icon on it and then you could press it and then Gemini pops up. And I'm just like, "Oh my God, Gemini's been here this whole time and I didn't even realize Gemini was always here." That's why I'm just like, "I don't know." - Jacqui C (AI Enthusiast)

While they express the intent to switch, reality is that browsers are very sticky so they may not end up following through on this intent



Our qual participants shared how onerous of task switching browsers is and that they already feel attached to their AI Tool of choice.

No, I think I have too many bookmarks on Chrome that I'd be sad to leave.
- Jack L (AI Enthusiast)

No, I'm just too tied to my browser.
- Amelia V (Cautious AI Optimist)

*I don't know if it'll become my primary tool , I'm just so used to it that **it's really hard for another AI chatbot to come between me and my ChatGPT.***
- Dee J (AI Enthusiast)



(click to play video, transcript in slide notes)

But some say they would be open to adopting it as a secondary browser to evaluate if the switch might be worthwhile after all.

I would just figure out, "Hey, this better, it's an AI browser. How much of my data is it getting? Is there a feature where I can turn off that?" I would go through that process. As great as Edge is, and it's great, I don't see myself as missing out on something. - Keith M (AI Enthusiast)

Switching between tabs and transferring context from one tab to another is a pain point

2 No Switching Between Tabs

Having to switch between tabs and transfer context from one tab to another is a pain point in need of a solution.

If it can take the information from across multiple tabs so that I don't have to extract all the bits and pieces that I'm explicitly looking for, that would save me 15, 20 minutes. – Peter S (AI Enthusiast)

I think it's a good idea because switching back and forth through different browsers is annoying. I have five different taps open and now I'm looking back to the Chat GPT to compare. – Jacqui C (AI Enthusiast)

When I access Copilot and Edge, I have to open a separate browser. I have to open a new tab essentially to access it, which is annoying because then it's not embedded into everything I'm already doing. It's a step away. – Rachel Z (AI Enthusiast)

I think it's really cool. I like how it's convenient. I always have to look up ChatGPT, press it on the browser, and then it takes me to a thing and then I have to open another tab to actually get to the conversation. I can see why they're trying to make it an easier way to navigate for people. – Amelia V (Cautious AI Optimist)

Eduardo D (AI Enthusiast)



(click to play video, transcript in slide notes)
Source: Qualitative Phase

This browser is appealing to me

71%
[Top 2 Box]

The TOOL that Makes Using AI Easy

This TOOL is built to provide the easiest way to use AI for your web tasks. It features an AI companion that is seamlessly integrated and has a flexible layout i and allows you to customize how it looks to suit your workflow. You won't need to switch between devices or navigate away from the page you're on

You can type or talk to your AI, It will also inform you when new capabilities and features

These same capability is also available on the browser for your mobile phone too.

15%+ 5-10% highlighted
Words Words = Positive
Words Words = Negative
Words Words = Polarizing

Q: Please rate this statement according to how much you agree or disagree. The browser is appealing to me (scenario Highlighter Tool)
N: AI Users (3,569)